

# ALU HUB TECHNICAL REPORT

ALU Hub is a flutter application that allows students looking for internship opportunities to connect with student-led startups.

## 1. Architecture:

ALU Hub uses frontend and backend architecture with flutter as the frontend and firebase as the backend. The application relies on several third party packages to deliver the functionalities:

- **flutter\_bloc** : For state management throughout alu hub application.
- **firebase\_core**: Required to connect application to firebase.
- **firebase\_auth**: Required in order to simplify the authentication process to firebase.
- **firebase\_storage**: This package handles the connection between alu hub and firebase storage to allow startups to upload their business registration certificates.
- **file\_picker**: This package allows flutter applications to use native os file explorer applications to upload files.
- **equatable**: This package enables us to compare objects in our code without writing repetitive code.
- **uuid**: This package simplifies the process of creating and using IDs which is especially useful for document based databases like firestore.

Communication Flow:

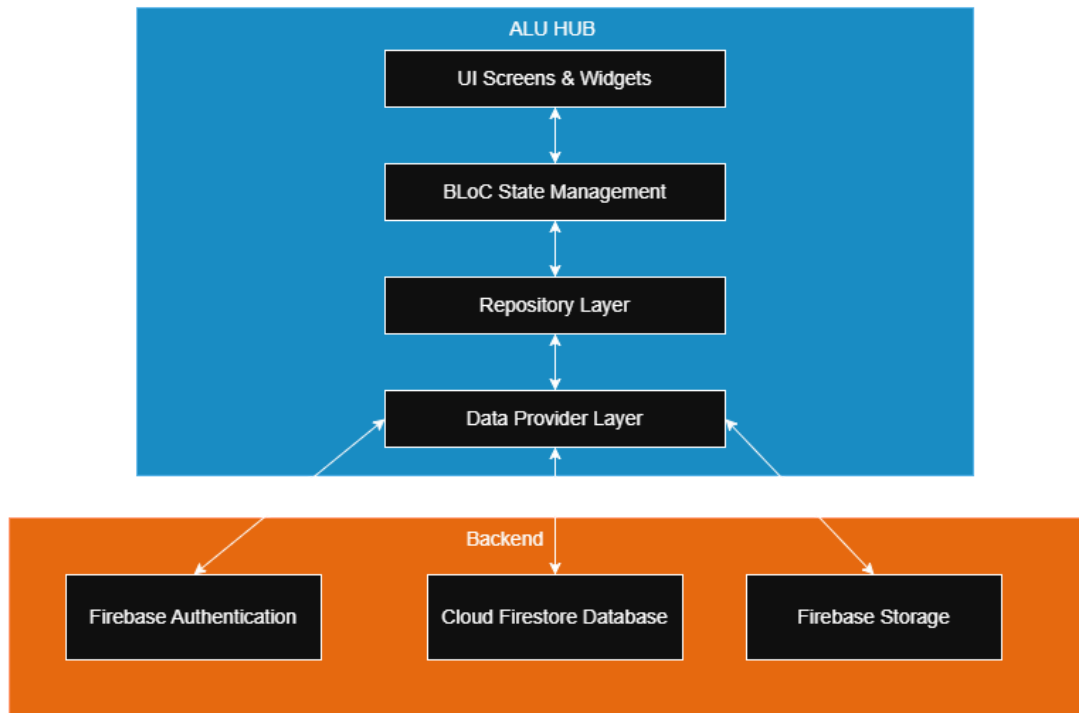
**UI** - This layer renders all the widgets and events

**BLoC** - This layer handles the overall application state, helping us to separate the UI from the business logic.

**Repository Layer** - This abstract layer contains code that determines whether to query firestore, upload to firebase storage, and invoke authentication.

**Data Provider Layer** - This layer communicates directly with firebase APIs.

Below is a diagram how the different layers interact with each other:



## 2. Workflow:

Here is a look at how students and startups will interact with the platform to connect and find the right opportunities.

## **Part 1: The Student Journey**

They begin by signing up and creating an account by filling in their profile information that will be seen by potential employers.



# ALU HUB

Bridging ALU Rwanda Students & Startups



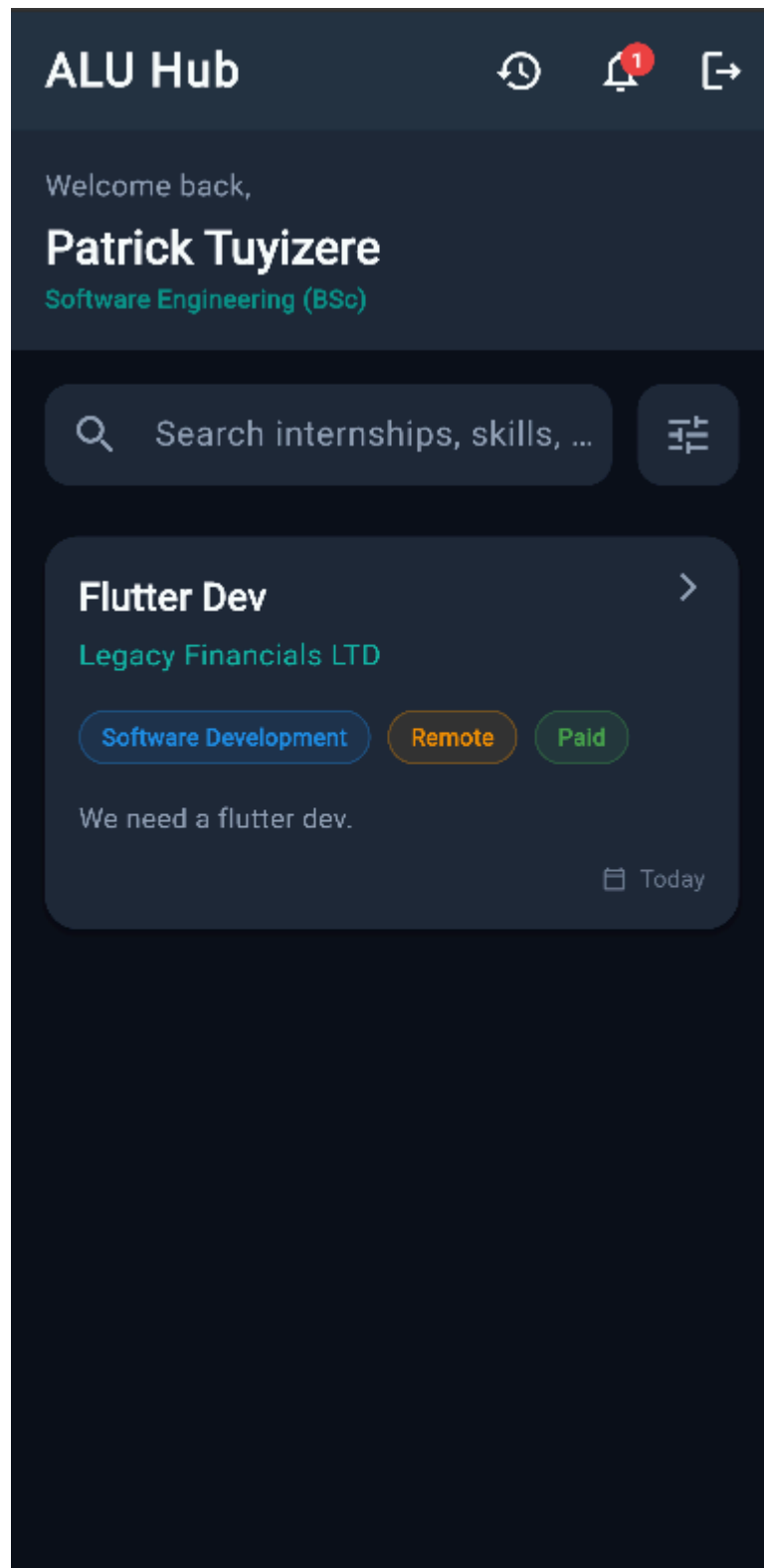
Login

NEW TO ALU HUB?

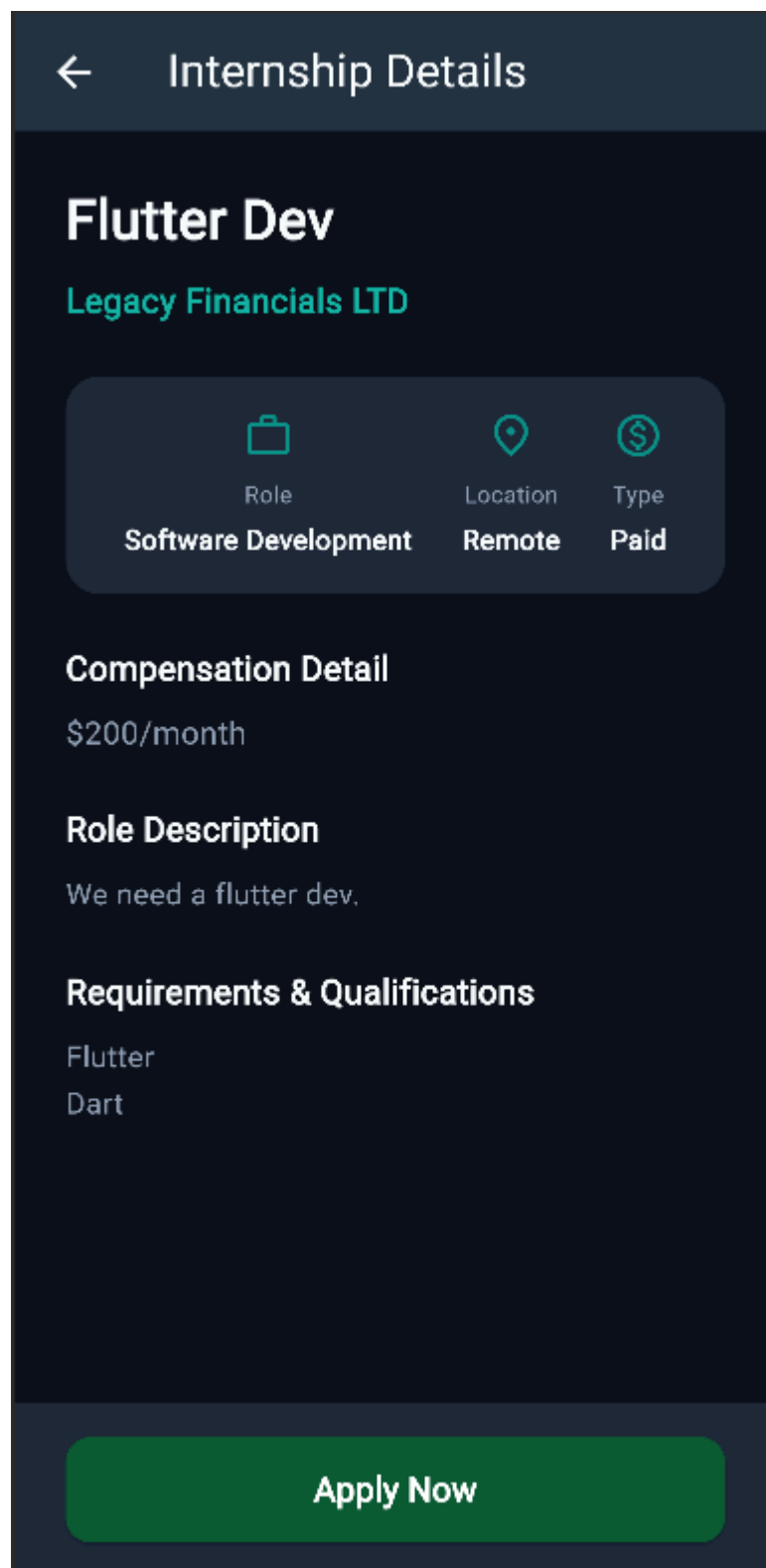
Register as Student

Register as Startup

Exploring Opportunities: Once their profile is set, they can login and explore available opportunities from their main dashboard, they can also search opportunities.



Applying: When a student finds an opportunity they like, they can click on it to view details and apply.



Making the Connection: Once they have applied and been accepted, they can see that in the notifications section which includes contact details of the startup.



## Notifications

### ● Application Approved!



Your application for "Flutter Dev" has been approved by Legacy Financials LTD. You can reach them directly via email at patk.tuy@gmail.com or by phone at 0794435673 to discuss the next steps!

#### Direct Contact Details:

 patk.tuy@gmail.com

 0794435673

13/7/2026 - 17:16

## Part 2: The Startup Journey

Signing Up: A company's journey on alu hub begins by setting up an account including uploading a copy of their RDB business registration.



Startup Registration



Contact Phone Number



Password





Website URL



Industry Sector



Company Description

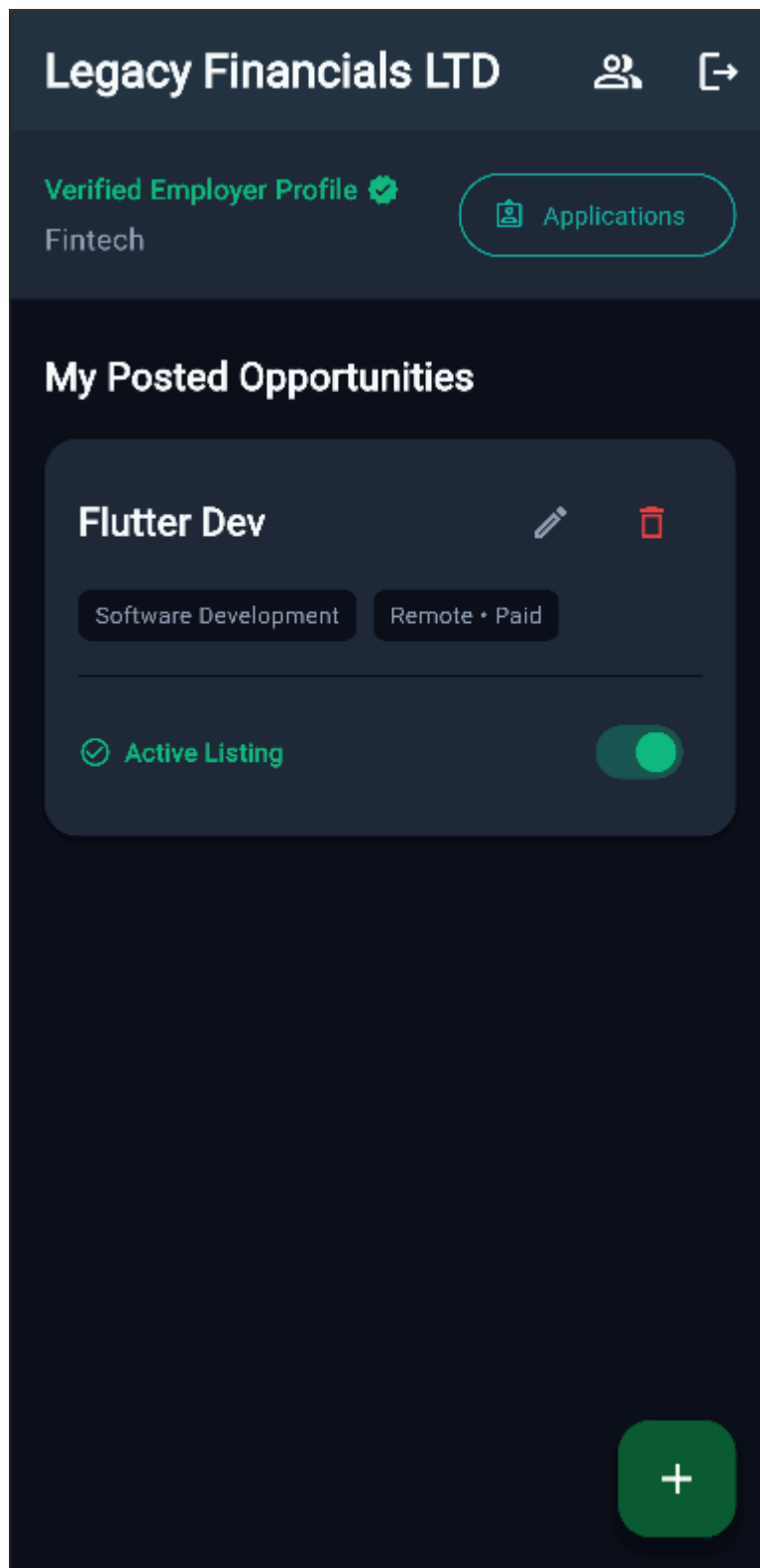


**RDB Registration Certificate**  
Upload Rwanda Development Board PDF registration certificate

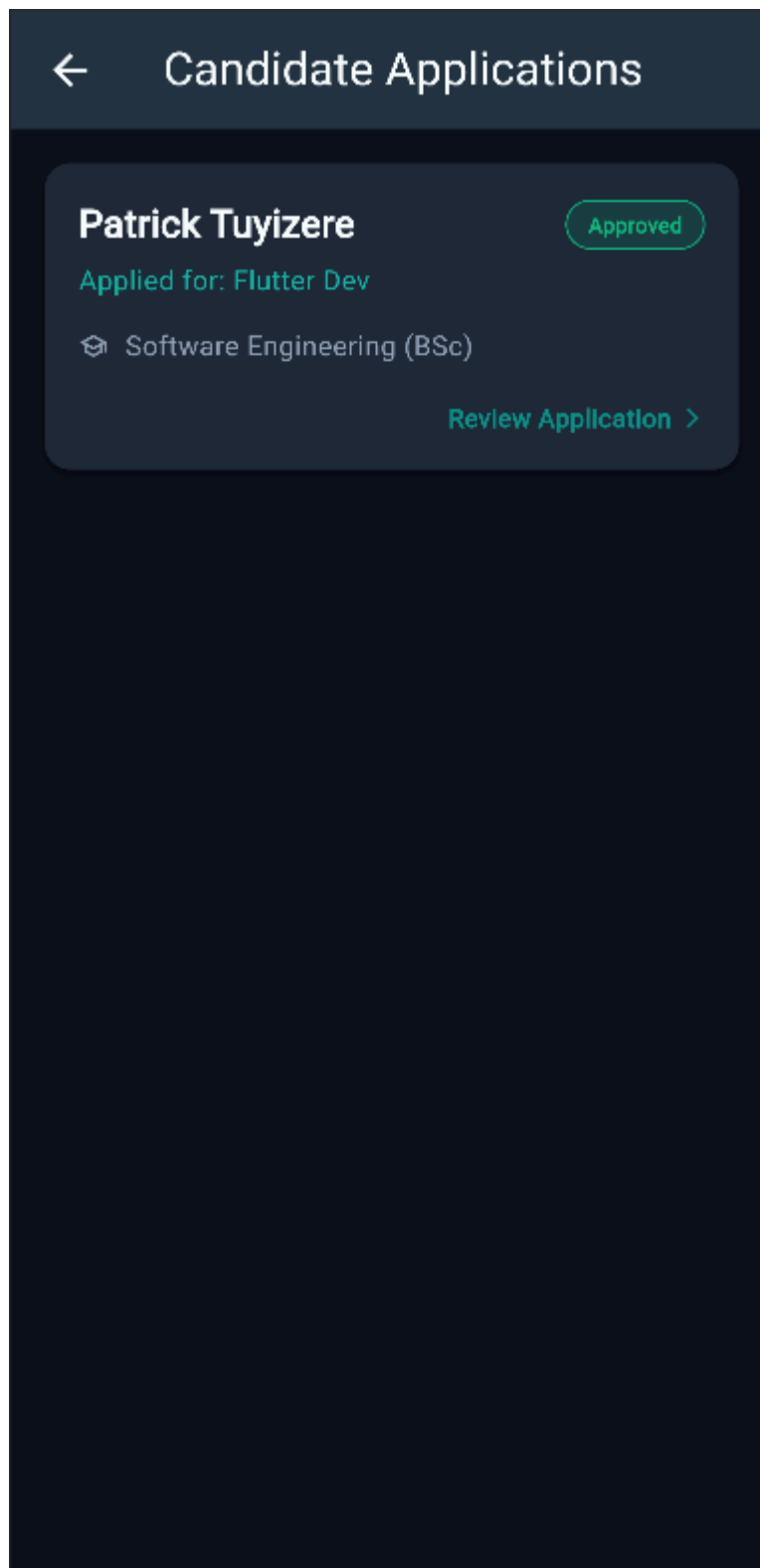
 Select PDF Certificate

Submit Profile for Verification

Verification: Once a business has been verified they get access to their account and are able to start posting internships.



Applicants: Once a candidate has applied to a posted opportunity, the startup can see their application from their applications screen.



Approving Candidate: Furthermore, the startup can approve or decline an application by clicking on the application to see its details and then deciding.

←

Review Application

Patrick Tuyizere

Software Engineering (BSc)

Position: Flutter Dev

Candidate Bio

I am a student who is passionate about software dev.

Skills & Core Competencies

Flutter

Dart

Figma

Java

Application Pitch / Cover Letter

I am very passionate about software development and I promise to do my best to make sure I am useful to your company. Thank you.

This application has been APPROVED

### **3. UI/UX Reasoning:**

The application takes a modern dark theme approach with emerald and teal accents while also heavily utilizing widgets like SingleChildScrollView, SafeArea to help avoid the notorious overflow errors that create yellow and black lines. The application also employs the use of simple and common widgets, this makes the user experience intuitive as opposed to using complex and rare widgets that might have a steep learning curve.

### **4. Scalability Considerations:**

The application is designed in a modular approach, separating the business logic from the user interface. Furthermore, data models are carefully split into the respective entities they represent and decoupled from one another. This is crucial because, should an entity require future modifications, the other entities need not change. By using a backend integrated service like firebase, we ensure that the application can grow with demand and be able to handle a large number of users should it be necessary.

### **5. Limitations & Future Improvements:**

No admin screen for self-registration: Admins have to manually verify startups via the database panel of Firebase Console. Future updates will provide a dedicated Admin web panel.

Static Resumes: Students enter text bios and skills list but do not upload resume PDFs Future iterations can include Resume file uploads using the same Firebase Storage mechanism.

Push Notifications: The app currently has in-app notification alerts that refresh dynamically. Add Firebase Cloud Messaging (FCM) to this you get system level push notifications .